

Two steps for virtual screening in the pocket of SARS-CoV-2 Mpro by MolAICal

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1. Introduction

In this tutorial, we introduce the fast way for drug virtual screening of SARS-CoV-2 Mpro based on the known database such as ZINC database. The premise for this tutorial is that you can deal with the receptor and ligand structures by MolAICal (see “Quick Start 1” in <https://molaical.github.io>). If users want to know more detailed molecular preparation process, you can learn from the tutorial: <https://github.com/MolAICal/documents/tree/master/tutorials/002-AIVS>. Here, MolAICal soft package is employed for this tutorial.

2. Materials

2.1. Software requirement

1) MolAICal: <https://molaical.github.io>

2.2. Example files

1) All the necessary tutorial files are downloaded from:

<https://github.com/MolAICal/tutorials/tree/master/003-VS>

2) The file named “ligandSet.mol2”, which contains 16 ligands obtained from the ZINC database, is chosen for the demo. You can select your ligand database.

3) The protein file named “pro.pdbqt” that is a PDBQT format structure of SARS-CoV-2 Mpro is used for molecular docking.

3. Procedure

Go to the tutorial directory:

```
#> cd 003-VS
```

3.1. Split ligands set into single molecules. If the drug database is built by single ligands with mol2 format, you can omit this step.

```
#> molaical.exe -tool mol2 -w split -n 1 -v true -m number -i ligandSet.mol2 -o splitdir
```

Note: when the second line content is valid in the mol2 file of each ligand, you can also use the second line string of the mol2 file as the file name by using the below command:

```
#> molaical.exe -tool mol2 -w split -n 1 -v true -i ligandSet.mol2 -o splitdir
```

Notice: Each molecular file in the docking set file, such as “ligandSet.mol2 must contain the full structure. If the molecular file in the docking set file, such as “ligandSet.mol2, **lacks hydrogen**, MolAICal may not generate the molecular file in PDBQT format. Users should first employ UCSF Chimera to add hydrogen to the molecule that lacks hydrogen, or use MolAICal to convert the molecule into PDBQT format by the following command (users should replace 1.mol2 with their own filename):

```
#> molaical.exe -tool format -i E:/1.mol2 -o E:/1.pdbqt
```

3.2. Configuring the file named “**conf.txt**” for docking (see the content below). The parameters

"receptor", center, and size should be assigned correctly according to the real receptors.

```
receptor = pro.pdbqt
out = all.pdbqt

cpu = 1
center_x = -10.733
center_y = 12.416
center_z = 68.829

size_x = 25
size_y = 30
size_z = 25

num_modes = 1
```

3.3. Running virtual screening. Before running the below command, please make sure the previous docking results are removed.

```
#> molaical.exe -dock vs -i splitdir -nc 3
```

Notice: If users use molecules in PDBQT format directly, users can close format conversion for virtual screening as below:

```
#> molaical.exe -dock vs -i splitdir -b off -k pdbqt -nc 3
```

-nc: represents the number of used CPU cores.

-b: switch molecular format conversion. "on" means conversion. "off" means no conversion. The default value is "on".

-k: the files contain the keyword for virtual screening. The default value is ".mol2".

4. Results

4.1 Checking results

Users can use Pymol software (<https://github.com/cgohlke/pymol-open-source-wheels> or <https://www.cgohlke.com>) to load ligands with PDBQT format directly. Here, UCSF Chimera is introduced to show docking results.

Open 003-VS/splitdir

1) Adding hydrogen (option)

```
#> molaical.exe -dock addh -i 1_out.pdbqt
```

2) Changing "pdbqt" to "pdb" format

```
#> molaical.exe -dock pdbqt2pdb -i 1_out.pdbqt
```

It will get a molecular file named “1_out.pdb”. And then use UCSF Chimera to load “1_out.pdb” and “protein.pdb”. The docking result is shown in Figure 1 that indicates MolAICal obtains the suitable ligand from the known molecular database.

Notice: I omit the surface show process in Figure 1. If users want to obtain the result like Figure 1, “Actions→Surface→show” should be selected in UCSF Chimera.

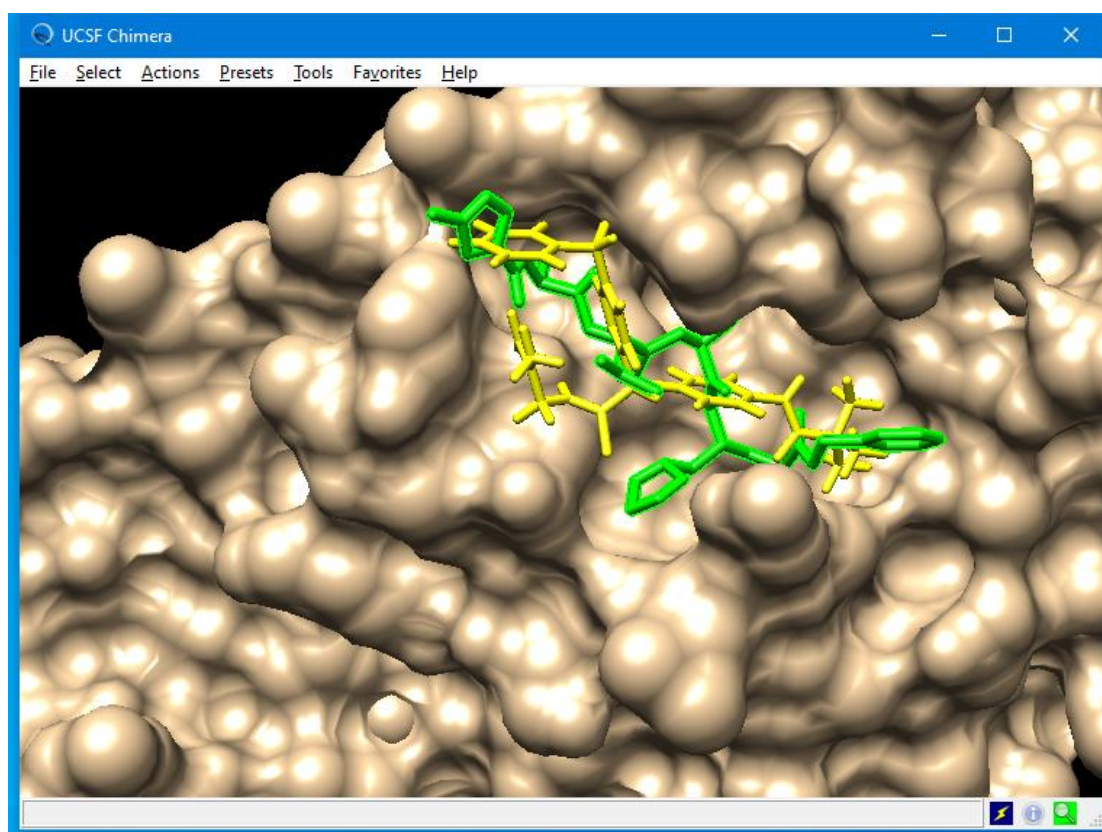


Figure 1. The ligand with green is the inhibitor N3 of SARS-CoV-2 Mpro. The ligand in yellow is obtained by virtual screening.

4.2 Ranking virtual screening results

Goto folder 003-VS, and run below command:

```
#> molaical.exe -call run -c sfile -i printscore.py "splitdir/*out.pdbqt"
```

It will show the ligand name and corresponding binding score (Sorted in ascending order by score). If users want to save results into a file, they can use below command:

```
#> molaical.exe -call run -c sfile -i printscore.py "splitdir/*out.pdbqt" > results.log
```

Note: “molaical.exe” is your installed directory path of MolAICal. All the useful scripts are supplied into the folder “scripts” of MolAICal.

4.3 Extract top-ranked molecule into the new folder

If users want to move top-ranked molecules into new folder, for example, in this tutorial, if users want to move 2 top-ranked molecules into new folder named “results”, they can use the command as follows:

```
#> molaical.exe -call run -c sfile -i get_top_results.py "splitdir/*out.pdbqt" 2 results
```

Two top-ranked molecules will be moved into folder “results”.

5. Continue VS (option)

Should the virtual screening task be terminated prematurely due to unforeseen circumstances, resuming the process requires only two steps:

- 1) Relocate all existing VS result files (containing the default “_out.pdbqt” suffix) to another directory by commands:

```
#> molaical.exe -call run -c sfile -i molaical_batch.py -sd splitdir -ds -ik "_out.pdbqt" > tmplist
```

-ds: Whether search files with appointed characters, if input “-ds” only, it means True to search files with appointed characters.

-sd: The directory for searching.

-ik: The keyword is for searching.

```
#> molaical.exe -call run -c sfile -i molaical_batch.py -cvs -il tmplist -ol l_complete_folder
```

-cvs: Whether prepare for continuing to run virtual screening. If input “-cvs” only, it means True to prepare for continuing to run virtual screening jobs.

-il: The file of ligands list.

-ol: In the section of continuing virtual screening when using “-cvs”, it is folder for storing.

Note: the complete molecules will be moved to the directory “l_complete_folder”.

Option: Sometimes, there may be huge numbers of molecules in one folder. To reduce the number of molecules in one folder, it can assign the number of molecules in one folder as follows:

```
#> molaical.exe -call run -c sfile -i molaical_batch.py -cvs -dp 3 -cvs -il tmplist -ol t1_folder
```

-dp: In the section of continuing virtual screening when using “-cvs”, it is the storing number of screened molecules in every folder. The default value is “None” which means to store all screened molecules in one folder.

It can be 1, 2, or 3... which represents the storing number of molecules in every folder.

Note: it will generate the folders "t1_folder1", "t1_folder2"... each of which contains the storing number after 'dp'.

2) Repeat the above virtual screening step to process unscreened molecules.

6. Linux Version of MolAICal: Architecture and Acceleration

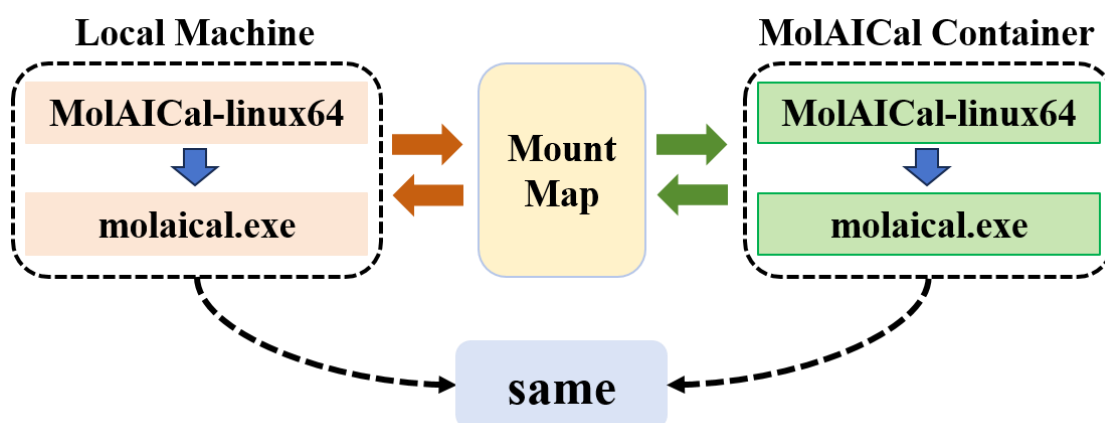


Figure 2. Architecture Diagram for the Linux Version of MolAICal

As shown in Figure 2, the **MolAICal container** can not only utilize the container system's files and programs but also access the host machine's library files through mounting and mapping mechanisms. This effectively **resolves** issues such as **library conflicts** or missing dependencies. However, **program execution within containers may indeed be slower than on the local machine.**

Figure 2 demonstrates that due to the mapping relationship, the computational commands executed by "**molaical.exe**" in the container are **identical** to those run by "**molaical.exe**" on the local machine.

◆ Therefore, **for tasks with minimal library dependencies**—such as virtual screening using MolAICal—**consider copying MolAICal from the container to the local machine:**

```
***** 1. Go to the container file system, it will enter the "/root" *****
#> molaical.exe -cset shell in

*** 2. The first part of 'cp' is from the MolAICal container, and the second part is in the local machine ***
#> cp -r soft/MolAICal-linux64 <local machine directory>/

***** 3. Exit container file system *****
#> exit
```

◆ At last, performing virtual screening locally via the same methodology can significantly accelerate the screening process.

Appendix

Other reference sorting methods:

If you work in Windows:

Open “results.log” by using Excel. In “Separator”, tick the space. Or you can copy all the content of “results.log” into Excel directly. Select all data in the second column and click “Sort Largest to Smallest” or “Sort Smallest to Largest” according to your needs (see Figure 3).

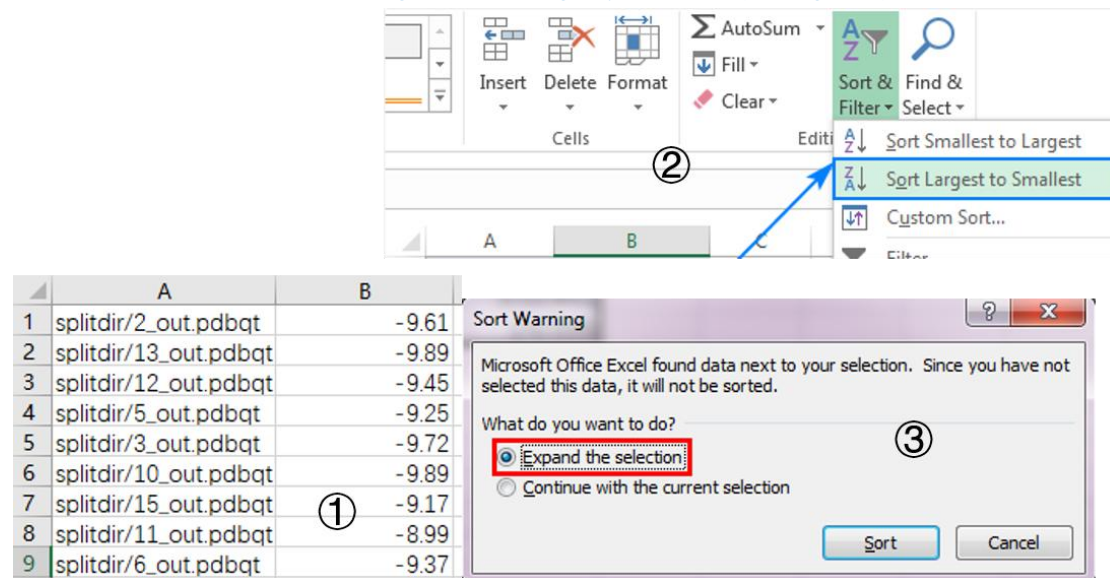


Figure 3. Ranking results

If you work in Linux, users can do it as below:

```
#> sort -n -t ' ' -k 2r results.log > rank.dat
```

Note: parameter “2r” is “Sort Smallest to Largest”, while “1r” is “Sort Largest to Smallest”.